CHAPTER 4

COMPLEX NUMBERS AND QUADRATIC EQUATIONS

CHAPTER ESSENCE AT A GLANCE:

Negative square roots used to break mathematics until mathematicians invented i = sqrt(-1)! On the 2D Argand plane, the real part sits on the X-axis and the imaginary part sits on the Y-axis. Every quadratic equation now has clean, complete solutions!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

- **Imaginary Unit Powers:** i = sqrt(-1), $i^2 = -1$, $i^3 = -i$, $i^4 = 1$. In general, divide exponent by 4: i^(4k + r) = i^r.
- **Complex Standard Form:** z = a + i b (where a = Re(z), b = Im(z)). Equality: $a_1 + i\,b_1 = a_2 + i\,b_2$ $a_1 = a_2$ and $b_1 = b_2$.
- **Conjugate & Modulus:** Conjugate z = a - i b. Modulus |z| = sqrt($a^2 + b^2$). Fundamental identity: $z \cdot z = |z|^2$.
- **Multiplicative Inverse:** z^{-1} = 1 over z = z over $|z|^2$ = (a - i b) over ($a^2 + b^2$). To simplify (a + ib) over (c + id), multiply top and bottom by (c - id).

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$z = x + iy \rightarrow |z| = \sqrt{x^2 + y^2}, \quad \bar{z} = x - iy$$

Modulus and Complex Conjugate Algebra

$$z \cdot \bar{z} = |z|^2 \rightarrow z^{-1} = \frac{\bar{z}}{|z|^2} \quad (z \neq 0)$$

Multiplicative Inverse via Modulus-Conjugate Relation

$$i = \sqrt{-1}, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1 \quad (i^{4k+r} = i^r)$$

Cyclic Powers of the Imaginary Unit

$$ax^2 + bx + c = 0 \rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (b^2 - 4ac < 0)$$

Complex Roots of Pure Quadratic Equations

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

- **Alert:** Watch the argument θ quadrant! If x < 0 and y > 0 (2nd quadrant), θ = π - α. If x < 0 and y < 0 (3rd quadrant), θ = -(π - α).
- **Alert:** To divide complex numbers (a + ib) over (c + id), simply multiply top and bottom by the conjugate (c - id)!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Solve the quadratic equation: $x^2 + x + 1 = 0$. [Score: 2 Marks]

Answer & Key Valuation Steps:

- **[Step 1: Quadratic Formula]** a = 1, b = 1, c = 1. Discriminant D = b^2 - 4 a c = 1^2 - 4(1)(1) = 1 - 4 = -3. [1 Mark]
- **[Step 2: Roots]** x = [-b ± √D] over (2 a) = [-1 ± √(-3)] over 2 = [-1 ± i √3] over 2. [1 Mark]
- **[Boxed Result: x = (-1 ± i √3) over 2]**

2. Solve the quadratic equation: sqrt(2) x^2 + x + sqrt(2) = 0. [Score: 2 Marks]

Answer & Key Valuation Steps:

- **[Step 1: Discriminant]** a = √2, b = 1, c = √2. D = b^2 - 4 a c = 1^2 - 4(√2)(√2) = 1 - 8 = -7. [1 Mark]
- **[Step 2: Roots]** x = [-1 ± √(-7)] over (2 √2) = [-1 ± i √7] over (2 √2). [1 Mark]
- **[Boxed Result: x = (-1 ± i √7) over (2 √2)]**

3. Express in the form a + i b: z = (1 + 2 i) over (1 - 3 i). [Score: 3 Marks]

Answer & Key Valuation Steps:

- **[Step 1: Multiplying Conjugate]** Multiply numerator and denominator by conjugate (1 + 3 i): z = [(1 + 2i)(1 + 3i)] over [(1 - 3i)(1 + 3i)]. [1 Mark]
- **[Step 2: Expansion]** Numerator = 1 + 3i + 2i + $6i^2$ = 1 + 5i - 6 = -5 + 5i. Denominator = 1^2 - $(3i)^2$ = 1 - 9(-1) = 1 + 9 = 10. [1 Mark]
- **[Step 3: Simplification]** z = (-5 + 5i) over 10 = -5 over 10 + 5 over 10 i = -1 over 2 + 1 over 2 i. [1 Mark]
- **[Boxed Result: z = -1 over 2 + 1 over 2 i]**

4. If x + i y = (a + i b) over (a - i b), prove that $x^2 + y^2 = 1$. [Score: 3 Marks]

Answer & Key Valuation Steps:

- **[Step 1: Taking Modulus on Both Sides]** |x + i y| = |(a + i b) over (a - i b)|. [1 Mark]
- **[Step 2: Modulus Properties]** |x + i y| = |a + i b| over |a - i b|. |x + i y| = √($x^2 + y^2$). |a + i b| = √($a^2 + b^2$). |a - i b| = √(a^2 + (-b)^2) = √(a^2 + b^2). [1 Mark]
- **[Step 3: Squaring]** √($x^2 + y^2$) = √($a^2 + b^2$) over √($a^2 + b^2$) = 1. Squaring both sides: $x^2 + y^2 = 1^2 = 1$. Hence proved. [1 Mark]
- **[Boxed Result: $x^2 + y^2 = 1$ proved]**

5. If $(x + i y)^3 = u + i v$, then show that u over x + v over y = 4 ($x^2 - y^2$). [Score: 3 Marks]

Answer & Key Valuation Steps:

- **[Step 1: Expansion of $(x + iy)^3$]** $(x + i y)^3 = x^3 + 3 x^2 (i y) + 3 x (i y)^2 + (i y)^3 = x^3 + 3 i x^2 y - 3 x y^2 - i y^3 = (x^3 - 3 x y^2) + i (3 x^2 y - y^3)$. [1.5 Marks]
- **[Step 2: Equating Real and Imaginary Parts]** $u = x^3 - 3 x y^2 = x (x^2 - 3 y^2) \Rightarrow$ u over x = $x^2 - 3 y^2$. $v = 3 x^2 y - y^3 = y (3 x^2 - y^2) \Rightarrow$ v over y = $3 x^2 - y^2$. [1 Mark]
- **[Step 3: Adding Both Expressions]** u over x + v over y = $(x^2 - 3 y^2) + (3 x^2 - y^2) = 4 x^2 - 4 y^2 = 4 (x^2 - y^2)$ = RHS. Hence proved. [0.5 Mark]
- **[Boxed Result: u over x + v over y = 4($x^2 - y^2$) proved]**

6. Find the modulus of: z = [(1 + i) over (1 - i)] - [(1 - i) over (1 + i)]. [Score: 4 Marks]

Answer & Key Valuation Steps:

- **[Step 1: Simplify First Term]** (1 + i) over (1 - i) = $(1 + i)^2$ over (1 - i^2) = 2i over 2 = i. [1.5 Marks]
- **[Step 2: Simplify Second Term]** (1 - i) over (1 + i) = $(1 - i)^2$ over (1 - i^2) = -2i over 2 = -i. [1.5 Marks]
- **[Step 3: Difference and Modulus]** z = i - (-i) = 2 i = 0 + 2 i. Modulus |z| = √($0^2 + 2^2$) = 2. [1 Mark]
- **[Boxed Result: z = 2 i, Modulus |z| = 2]**

7. (a) If |z1| = |z2| = 1, prove that |z1 + z2| = |1 over z1 + 1 over z2|. (2 Marks)n(b) If (a + i b)(c + i d)(e + i f)(g + i h) = A + i B, prove that $(a^2 + b^2)(c^2 + d^2)(e^2 + f^2)(g^2 + h^2) = A^2 + B^2$. (2 Marks) **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• **[Step 1: Modulus Property for Reciprocals]** Since $|z| = 1 \Rightarrow |z|^2 = z \cdot z_{bar} = 1 \Rightarrow 1 \text{ over } z = z_{bar}$. 1 over z1 + 1 over z2 = z1 bar + z2 bar = (z1 + z2) bar. Taking modulus: |1 over z1 + 1 over z2| = |(z1 + z2) bar| = |z1 + z2|. Hence proved. [2 Marks]

• **[Step 2: Product Modulus Proof]** Taking modulus on both sides: |a + ib| |c + id| |e + if| |g + ih| = |A + iB|. $\sqrt{(a^2 + b^2)}\ \sqrt{(c^2 + d^2)}\ \sqrt{(e^2 + f^2)}\ \sqrt{(g^2 + h^2)} = \sqrt{(A^2 + B^2)}$. Squaring both sides: $(a^2 + b^2)(c^2 + d^2)(e^2 + f^2)(g^2 + h^2) = A^2 + B^2$. Hence proved. [2 Marks]

• **[Boxed Result: Both identities proved]**